Fields respond to temperature much as ordinary matter does. The higher the temperature, the more ferociously the value of a field will—like the surface of a rapidly boiling pot of water—undulate up and down. At the chilling temperature characteristic of deep space today (2.7 degrees above absolute zero, or 2.7 Kelvin, as it is usually denoted), or even at the warmer temperatures here on earth, field undulations are minuscule. But the temperature just after the big bang was so enormous—at 10^{-43} seconds after the bang, the temperature is believed to have been about 10^{32} Kelvin—that all fields violently heaved to and fro.

As the universe expanded and cooled, the initially huge density of matter and radiation steadily dropped, the vast expanse of the universe became ever emptier, and field undulations became ever more subdued. For most fields this meant that their values, on average, got closer to zero. At some moment, the value of a particular field might jitter slightly above zero (a peak) and a moment later it might dip slightly below zero (a trough), but on average the value of most fields closed in on zero—the value we intuitively associate with absence or emptiness.

Here's where the Higgs field comes in. It's a variety of field, researchers have come to realize, that had properties similar to other fields' at the scorchingly high temperatures just after the big bang: it fluctuated wildly up and down. But researchers believe that (just as steam condenses into liquid water when its temperature drops sufficiently) when the temperature of the universe dropped sufficiently, the Higgs field condensed into a particular *nonzero* value throughout all of space. Physicists refer to this as the formation of a *nonzero Higgs field vacuum expectation value*—but to ease the technical jargon, I'll refer to this as the formation of a *Higgs ocean.*

It's kind of like what would happen if you were to drop a frog into a hot metal bowl, as in Figure 9.1a, with a pile of worms lying in the center. At first, the frog would jump this way and that—high up, low down, left, right—in a desperate attempt to avoid burning its legs, and on average would stay so far from the worms that it wouldn't even know they were there. But as the bowl cooled, the frog would calm itself, would hardly jump at all, and, instead, would gently slide down to the most restful spot at the bowl's bottom. There, having closed in on the